**file_reader.c** - the file reader we used previously
**custom_reader.c** - we specify the amount of bytes we want to read at the end of command, same file_reader logic
**governor_controller.c** - while reading a file, this program in short time intervals detects if the file is read from the disk or pagecache and in case of disk, it changes the governor to "powersafe"

Step 1. *build the programs*
gcc -O0 file_reader.c -o file_reader
gcc -O0 custom_reader.c -o custom_reader
gcc -O0 governor_controller.c -o governor_controller

Step 2. *generate the data.bin*
dd if=/dev/urandom of=data.bin bs=1M count=1000 status=progress

Step 3. *clean the page cache*
echo 3 | sudo tee /proc/sys/vm/drop_caches > /dev/null

Step 4. *check the governor in the other window of terminal*
watch -n 0.2 "cat /sys/devices/system/cpu/cpufreq/policy0/scaling_governor"

Step 5. *run the program*

sudo ./governor_controller 50 3 5 ./file_reader data.bin
or
sudo ./governor_controller 50 3 5 ./custom_reader data.bin N

where
50 → sampling interval in milliseconds
3 → number of consecutive DISK windows needed before switching to powersave
5 → number of consecutive CACHE windows needed before restoring the old governor

```
mataghinim@mataghinim-ThinkPad-L460:~/Desktop/capstonenew/gov_change$ sudo ./governor_controller 50 3
5 ./file_reader data.bin
Started reader PID=22309
window=DISK drchar=20975612 dread_bytes=23359488 disk_streak=1 cache_streak=0
window=DISK drchar=25165824 dread_bytes=25165824 disk_streak=2 cache_streak=0
[ACTION] switched to powersave after 3 DISK windows
window=DISK drchar=25165824 dread_bytes=25165824 disk_streak=3 cache_streak=0
window=DISK drchar=33554432 dread_bytes=33554432 disk_streak=4 cache_streak=0
window=DISK drchar=25165824 dread_bytes=25165824 disk_streak=5 cache_streak=0
window=DISK drchar=25165824 dread_bytes=25165824 disk_streak=6 cache_streak=0
```

.

.

.

```
window=DISK drchar=25165824 dread_bytes=25165824 disk_streak=35 cache_streak=0
window=DISK drchar=25165824 dread_bytes=25165824 disk_streak=36 cache_streak=0
window=DISK drchar=25165824 dread_bytes=25165824 disk_streak=37 cache_streak=0
window=DISK drchar=25165824 dread_bytes=25165824 disk_streak=38 cache_streak=0
window=DISK drchar=25165824 dread_bytes=24117248 disk_streak=39 cache_streak=0
window=DISK drchar=25165824 dread_bytes=25165824 disk_streak=40 cache_streak=0
window=DISK drchar=25165824 dread_bytes=25165824 disk_streak=41 cache_streak=0
Read 1000.00 MB in 2.113 s (473.22 MB/s)
Reader exited.
```

Here we see how, after 2-3 intervals of DISK-bound read, the program changes the governor to powersave.

```
mataghinim@mataghinim-ThinkPad-L460:~/Desktop/capstonenew/gov_change$ sudo ./governor_controller 50 3
5 ./file_reader data.bin
Started reader PID=26335
window=CACHE drchar=408948732 dread_bytes=0 disk_streak=0 cache_streak=1
window=CACHE drchar=420478976 dread_bytes=0 disk_streak=0 cache_streak=2
Read 1000.00 MB in 0.126 s (7959.80 MB/s)
Reader exited.
```

Also, we see how when we do not clean and run the same thing again, the page caches are already full of content, so we take from there and do not touch the governor (yet)